

Particulate Matter Monitoring & Health Effects

Issue 2026-08-01 | Entry window 31 July–1 August 2026 | 15 records in scope

15RECORDS IN SCOPE
(8 SUBTOPICS)**10**EFFECT ESTIMATES
EXTRACTED**5**RECORDS FROM CHINA
(33% OF CORPUS)**3**REPRODUCTIVE /
DEVELOPMENTAL**0**SENSING /
INSTRUMENTATION

Signal of the day

Road-network proxies fail against personal $\text{PM}_{2.5}$ in sub-Saharan Africa — and they fail with the wrong sign, not merely weakly.

Nyoni et al. (*Air Qual Atmos Health*) validated three of the exposure proxies that epidemiology routinely substitutes for measurement — population-weighted road network density, Euclidean distance to highway, Euclidean distance to main road — against **a year of filter-based personal $\text{PM}_{2.5}$ from 343 postpartum participants** in The Gambia, Kenya and Mozambique. Weighted road density correlated with personal $\text{PM}_{2.5}$ only in Mozambique ($\rho = 0.351$, $p = 0.005$); Kenya and The Gambia were null ($\rho = -0.041$ and -0.020). Distance-to-highway was *positive* in The Gambia and Mozambique but **significantly negative in Kenya** ($\rho = -0.224$, $p = 0.018$) — the proxy inverts across countries. Single-variable random forests never exceeded $R^2 = 0.45$; combining all three reached $R^2 = 0.52$ (RMSE $31.7 \mu\text{g}/\text{m}^3$) in Kenya and 0.60 ($8.9 \mu\text{g}/\text{m}^3$) in Mozambique, with a global model at 0.46 . An RMSE of $30 \mu\text{g}/\text{m}^3$ is larger than the entire annual mean in most European cohorts. The operational reading for distributed sensing is direct: in settings where household energy and biomass burning dominate the personal exposure budget, traffic geometry is not a transferable surrogate, and the case for physically deploying low-cost nodes rather than regressing on road maps is strongest exactly where reference infrastructure is thinnest.

What changed today

- **The window is reproductive-heavy and mechanistically thin.** Three of fifteen records address preconception or in-utero exposure (cleft lip/palate, birth weight, small-for-gestational-age), and all three take the same methodological posture — mixture models or vulnerability stratification rather than single-pollutant point estimates. None used measured personal exposure; all assigned exposure from monitors or models to a residential address.
- **Effect heterogeneity is being reported as the finding, not the caveat.** Bello et al. find essentially no average prenatal $\text{PM}_{2.5}$ effect on birth outcomes in low-pollution Quebec but a **6.6 g birth-weight decrement per $1 \mu\text{g}/\text{m}^3$** in the most socioeconomically vulnerable stratum; Chen et al. find their 29-hazard mixture associated with asthma **only** in high-social-vulnerability census tracts (OR 1.35, 95% CI 1.06–1.73 per quintile). Population-average nulls are increasingly shown to conceal stratum-specific signals, which raises the sampling requirement for any monitoring network intended to support this class of inference.
- **Component-resolved $\text{PM}_{2.5}$ keeps out-performing mass.** Zhang et al. (Henan Rural Cohort, $n = 14,379$) report black carbon carrying **54–86% of the WQS weight** for cognitive dysfunction across four outcome definitions, organic matter second. This is now a recurring result in this digest rather than an isolated one, and it is a direct argument for absorption- or thermal-optical channels alongside nephelometric mass in distributed nodes.
- **Zero sensing or instrumentation records for the second consecutive window.** This is a harvest defect, not a real absence in the field: the pipeline still has no date-windowed channel into *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.* See the provenance box.

Corpus analytics

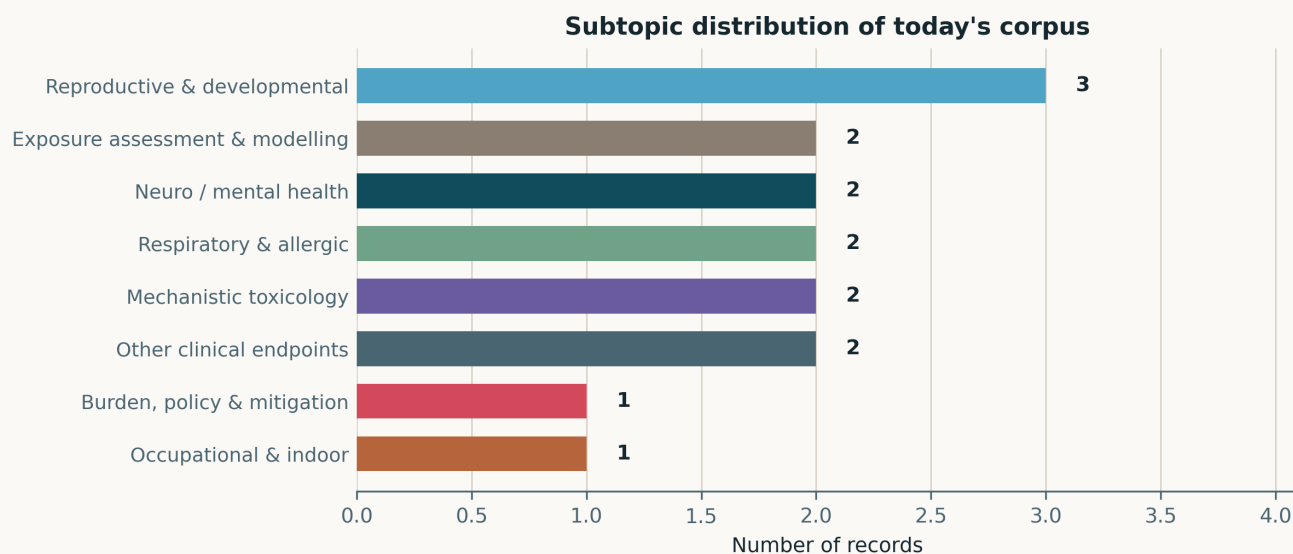


Fig. Subtopic assignment of the 15 in-scope records. Assignment is single-label by dominant contribution. Reproductive & developmental leads at 3 records; two subtopics present in earlier issues — Cardiovascular & metabolic and Sensing, forecasting & instrumentation — are empty this window. Colours are fixed across issues so a subtopic keeps its hue in the weekly rollup.

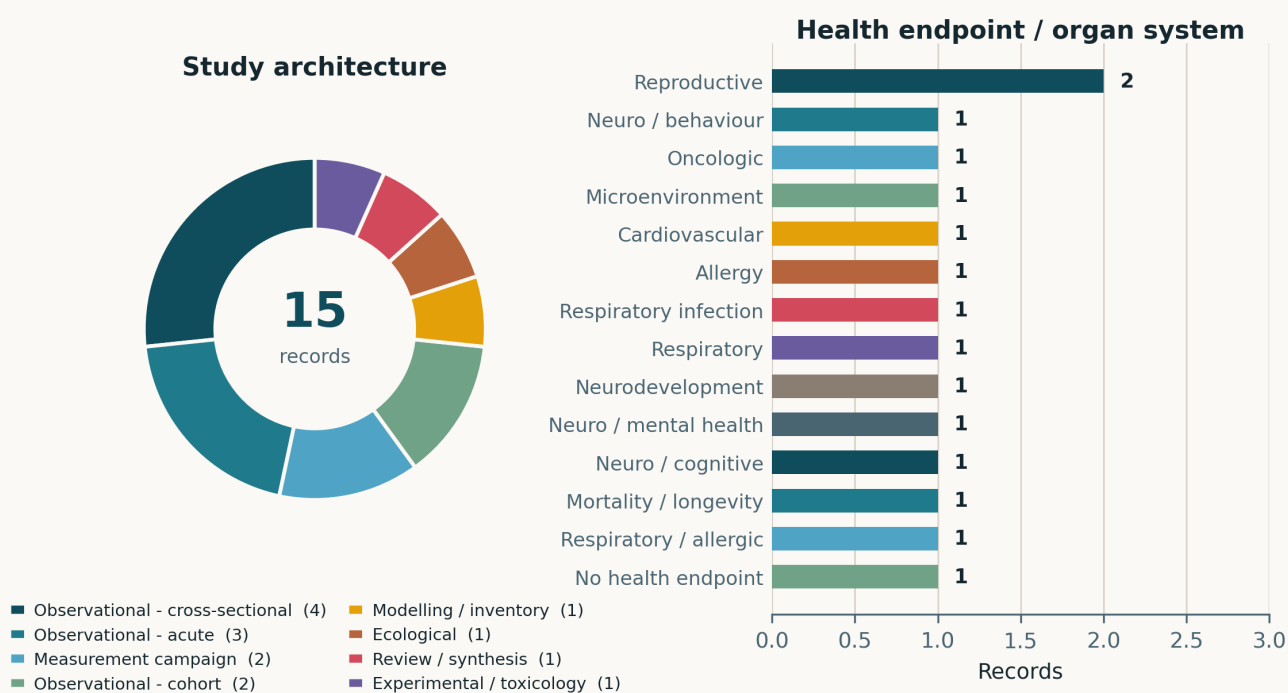


Fig. Left — study architecture. Observational designs account for 9/15 (60%), split 4 cross-sectional (including two case-control and one exposome-wide analysis), 3 acute time-series or panel, and 2 cohort. Only one experimental-toxicology record and one evidence synthesis appear. Right — health endpoint. Two records carry no human health outcome: the exposure-proxy validation (plotted as *No health endpoint*) and the indoor-air characterisation (plotted as *Microenvironment*).

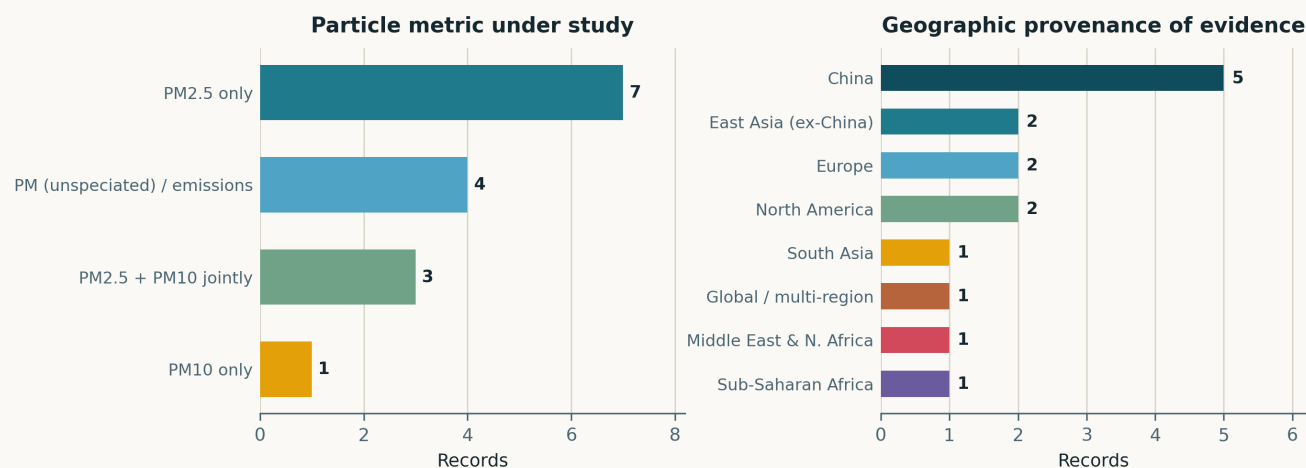


Fig. Left — particle metric. PM_{2.5} alone is the exposure in 7 records, PM_{2.5} and PM₁₀ jointly in 3, PM₁₀ alone in 1, and 4 records use an unspiciated or non-ambient particulate metric (diesel SRM 1650b, indoor incense PM, indoor construction PM, traffic gases standing in for PM). Right — geography. China supplies 5 records (33%); the single sub-Saharan African record is the exposure-proxy validation study.

Life-course windows addressed today (records may span several stages)

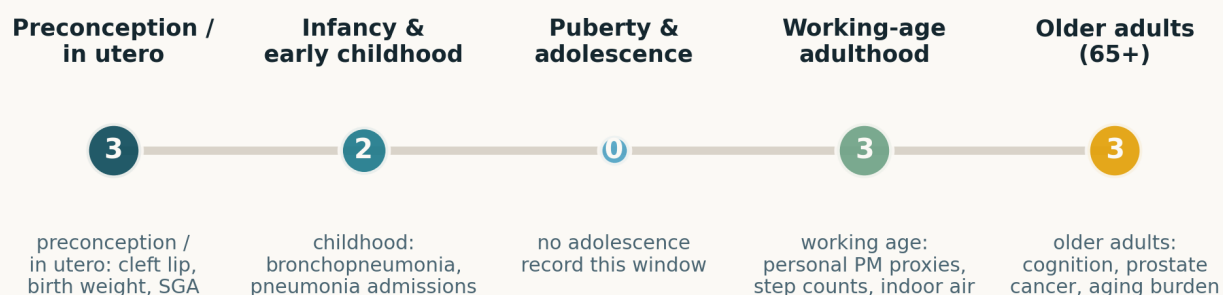


Fig. Life-course windows addressed. Counts are non-exclusive. The preconception and in-utero column is the largest this window, matched by working-age and older-adult; no record addresses adolescence. Records with no identifiable human life-course window — the ex-vivo cardiac preparation and the rodent meta-analysis — are excluded from the tally.

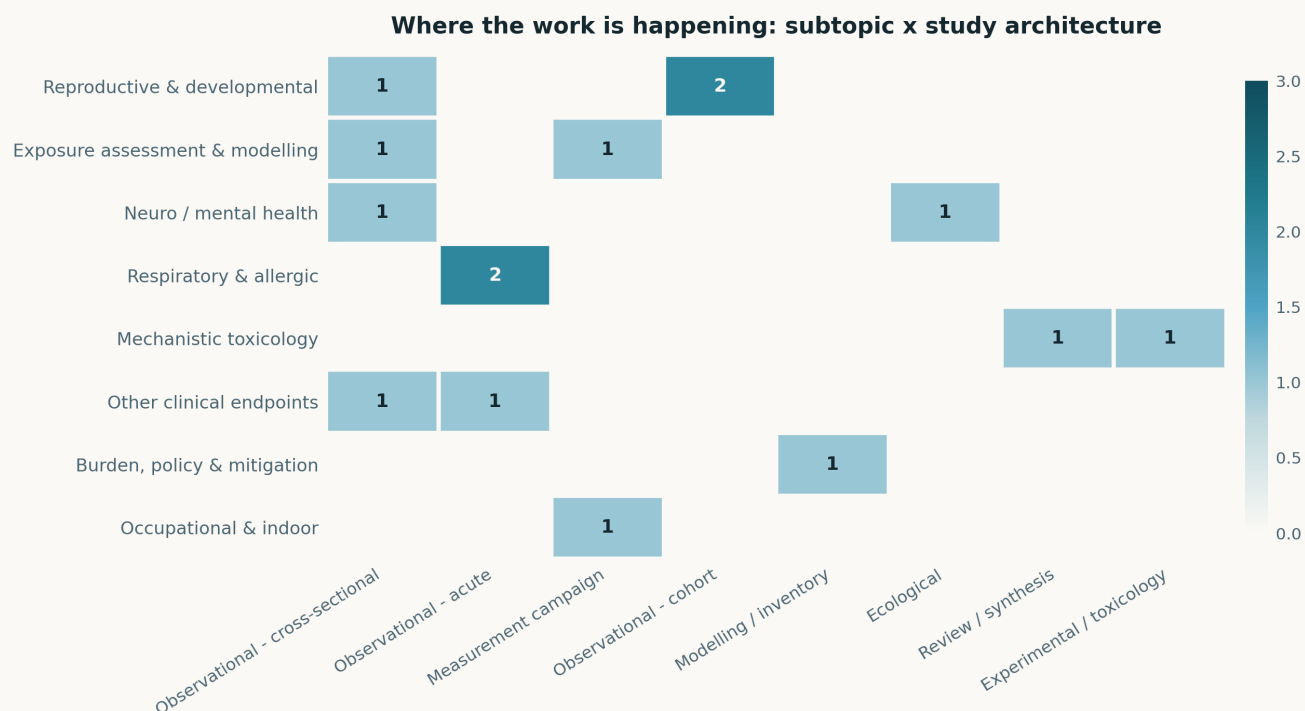


Fig. Subtopic × study-architecture cross-tabulation. The structural feature to note is the empty Sensing row and the concentration of reproductive and exposure work in observational cells — no experimental or measurement-campaign record supports the reproductive findings, so every exposure–outcome pair in that band rests on modelled or monitor-assigned exposure.

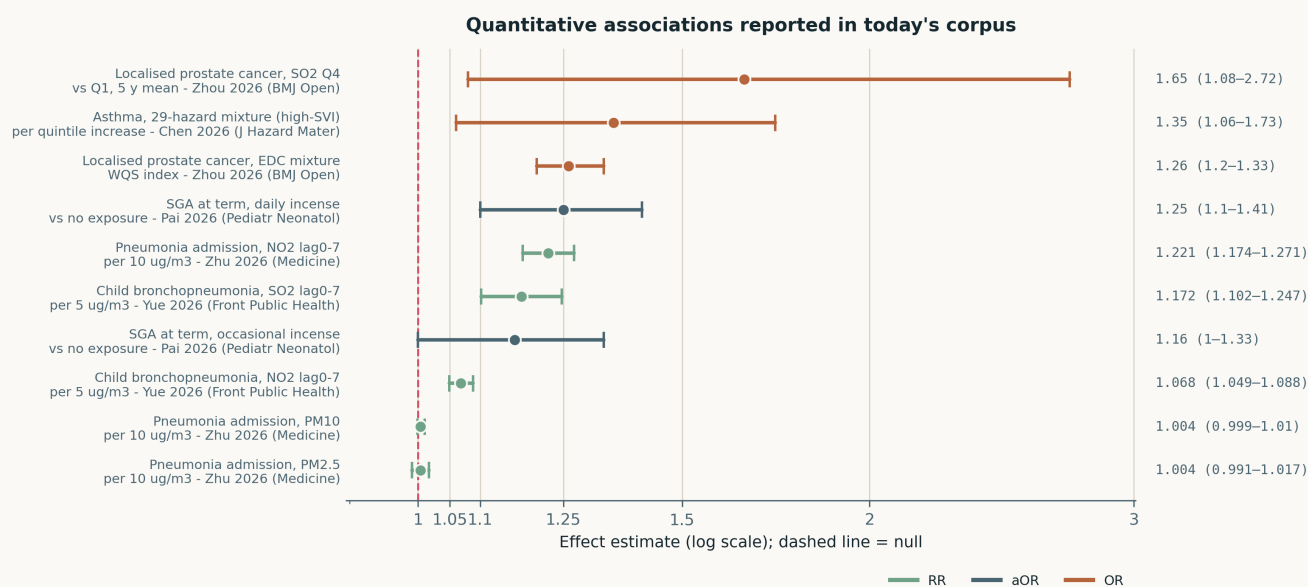


Fig. All ten quantitative effect estimates in the corpus, on a common ratio axis. The exposure contrasts are not commensurable and the plot must not be read as a meta-analysis: increments range from per 5 $\mu\text{g}/\text{m}^3$ to per 10 $\mu\text{g}/\text{m}^3$ to quartile and quintile contrasts to a WQS index, and the outcome scale mixes RR, OR and adjusted OR. Note in particular that the two PM-specific estimates in the set (Zhu et al., PM_{2.5} and PM₁₀ pneumonia admissions) are the *smallest* and are null or borderline, while the large estimates all belong to gaseous co-pollutants or mixture indices — a reminder that in multi-pollutant time-series the PM term is often the weakest term in its own model.

Paper-level digest

Reproductive & developmental

3 records

Bello et al. 2026 Econ Hum Biol

Prenatal exposure to PM_{2.5} and infant health: evidence from Quebec

Administrative birth records for Quebec 2008–2015 linked to air-quality monitoring and to tax records, in neighbourhood-by-quarter fixed-effects models. At the population level prenatal PM_{2.5} has **little effect on birth outcomes** in this low-pollution setting. The result is in the heterogeneity: within the most socioeconomically vulnerable families, a **1 µg/m³ increase in mean prenatal PM_{2.5} is associated with ≈6.6 g lower birth weight**, and days exceeding the Canadian 24-h standard carry disproportionate effect beyond the mean. Effects on prematurity and birth weight are also detectable among newborns admitted to neonatal care. The threat to inference is exposure misclassification: monitor-to-residence assignment in a low-gradient setting compresses the true contrast, and the vulnerability index is constructed post hoc from tax data. Relevant to distributed sensing because the effect the authors recover lives in short exposure peaks — precisely the temporal structure that sparse regulatory networks alias away.

doi: 10.1016/j.ehb.2026.101631

Pai et al. 2026 Pediatr Neonatol

Household incense burning and small for gestational age risk in newborn infants

18,666 mother–infant pairs from the Taiwan Birth Cohort Study, incense exposure during pregnancy self-reported as none, occasional or daily, outcome SGA stratified by term and preterm. SGA prevalence rose monotonically across exposure categories (8.1%, 9.4%, 10.0%). Among term infants, **aOR 1.16 (95% CI 1.00–1.33) for occasional and 1.25 (1.10–1.41) for daily exposure**, robust to adjustment for other indoor sources, gestational complications and weight gain, and to interaction with secondhand smoke. Preterm strata showed the same direction without significance; no association with preterm birth itself. The limitation is that exposure is a three-level questionnaire item with no measured indoor PM concentration, duration or ventilation covariate, so the dose axis is ordinal at best, and residual confounding by household characteristics linked to religious practice is not excluded. A well-powered demonstration of how much epidemiological signal is currently carried by a categorical proxy that a single indoor node would replace with a real concentration.

doi: 10.1016/j.pedneo.2026.06.003

Sezer et al. 2026 Birth Defects Res

Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate

Hospital-based case-control study, 127 nonsyndromic CLP cases against 271 controls, with PM_{2.5}, PM₁₀, O₃, NO_x, SO₂ and CO assigned for the three months before conception and the first trimester. Higher O₃, PM_{2.5} and PM₁₀ were each associated with elevated CLP risk in single-pollutant models; **O₃ was the dominant contributor in the WQS mixture** and the only exposure stable across windows and model specifications, while traffic-related pollutants did not survive multi-pollutant adjustment. The study is underpowered for a mixture analysis — 127 cases against a six-pollutant WQS index invites weight instability, and no bootstrap weight distribution is reported — and hospital-based control selection risks referral bias. Read as hypothesis-generating. The methodological point that carries: with correlated pollutants and a small case series, the pollutant that wins the mixture is often the one with the widest exposure contrast, not the one with the strongest biology.

doi: 10.1002/bdr2.70102

Exposure assessment & modelling

2 records

Nyoni et al. 2026 Air Qual Atmos Health

Performance of road-traffic-based exposure proxies against personal PM_{2.5} measurements in three sub-Saharan African countries

One year of filter-based outdoor personal PM_{2.5} from 343 postpartum participants in The Gambia, Kenya and Mozambique, used to validate population-weighted road network density (wRND), Euclidean distance to highway (EH) and to main road (EM). Spearman correlations were weak and **sign-inconsistent across countries**: wRND significant only in Mozambique ($\rho = 0.351$), EH positive in The Gambia (0.335) and Mozambique (0.292) but **negative in Kenya** (-0.224 , $p = 0.018$). Single-proxy random forests gave $R^2 < 0.45$ everywhere; combined models reached $R^2 = 0.52$ (RMSE 31.7 µg/m³) in Kenya and 0.60 (8.9 µg/m³) in Mozambique. The sampled population — postpartum women — is not representative of general mobility patterns, and only outdoor-filtered personal exposure was measured, so the household cooking contribution that most plausibly drives the proxy failure is inferred rather than partitioned. Still the strongest available empirical case that road-geometry proxies require local validation before use, and a direct argument for node deployment over GIS surrogacy.

doi: 10.1007/s11869-026-02060-y

Chen et al. 2026 J Hazard Mater

Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability

Exposome-wide association study over 29 residential hazards — air pollutants, industrial VOC emissions, highway density and proximity to 13 pollution-source classes — in 7,092 North Carolina adults (2013–2022), stratified at the median census-tract Social Vulnerability Index, with quantile g-computation for the mixture. In the overall population only proximity to dry-cleaning-solvent-contaminated sites was associated with asthma. In high-SVI tracts, **eight hazards — predominantly PM_{2.5} constituents and VOCs — were positively associated at FDR < 0.05**, and each one-quintile increase in the hazard mixture carried **35% higher asthma odds (95% CI 6–73%)**. Toluene, dry-cleaning proximity, PM organic matter and carbon monoxide each contributed over 10% of the joint association. Hazard correlations were themselves stronger in high-SVI areas, which is both the substantive finding and the analytical problem: with collinearity differing by stratum, the g-computation weights are not directly comparable across strata. Cross-sectional, self-reported physician-diagnosed asthma, residence-only exposure.

doi: 10.1016/j.jhazmat.2026.143083

Neuro, cognition & mental health

2 records

Zhang et al. 2026 Neurotoxicology

Association of PM_{2.5} and its components with cognitive dysfunction among rural older adults

14,379 participants aged ≥60 in the Henan Rural Cohort, cognition by MMSE and the Hasegawa Dementia Scale, PM_{2.5} and its black carbon, organic matter, ammonium, sulfate and nitrate fractions from the TAP reanalysis, analysed by logistic and weighted quantile sum regression.

Each IQR increase in PM_{2.5} or any component raised the odds of cognitive dysfunction by **19–43%**, and **black carbon carried 86.4% to 54.3% of the WQS weight** across the four outcome definitions, organic matter 7.2% to 34.6%. A network-toxicology layer nominated 47 candidate genes with RPL23, RPL36AL and RPS25 prioritised by random forest, implicating ribosomal and mitochondrial pathways. Two limitations bound the reading: the design is cross-sectional, so reverse causation through mobility and residential choice in an ageing rural cohort is live; and the component estimates come from a 1 km reanalysis whose component-level errors are spatially correlated with total mass, which inflates apparent component separation. The in-silico gene prioritisation is hypothesis-generating only.

doi: [10.1016/j.j.neuro.2026.103539](https://doi.org/10.1016/j.j.neuro.2026.103539)

Daoud et al. 2026 Environ Sci Pollut Res Int

Air pollution and suicide mortality in the UK: implications for environmental and mental health policy

Balanced panel of over 2,200 local-authority-year observations across the UK 2015–2023, ONS suicide counts against DEFRA modelled PM_{2.5}, in two-way fixed-effects regression. The cross-sectional association is *negative*; after local-authority and year fixed effects it turns positive ($\beta = 0.221$, $p = 0.007$), with a nonlinear specification indicating a U-shape and a turning point near **7.8 $\mu\text{g}/\text{m}^3$** . The sign reversal on adding fixed effects is the whole result and it is fragile: no time-varying socioeconomic controls (deprivation, unemployment, service provision) are included, and those are exactly the confounders that move with both PM_{2.5} and suicide within a local authority over nine years. Ecological design, modelled rather than measured exposure, area-level aggregation of an individual-level outcome. The authors label the findings exploratory and that label should be honoured. Included for completeness of the mental-health signal, not as evidence.

doi: [10.1007/s11356-026-38089-w](https://doi.org/10.1007/s11356-026-38089-w)

Respiratory & allergic

2 records

Yue et al. 2026 Front Public Health

Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou

20,623 paediatric bronchopneumonia admissions (ICD-10 J18, ages 0–14) at a single tertiary hospital in Huadu District, Guangzhou, 2014–2023, against daily NO₂ and SO₂ in a quasi-Poisson generalised additive model. Per 5 $\mu\text{g}/\text{m}^3$, peak single-day excess risk at lag 4 was **2.74% (95% CI 1.73–3.75) for NO₂** and **8.81% (4.94–12.82) for SO₂**; over the 0–7-day moving average these rose to **6.83% (4.93–8.77)** and **17.21% (10.18–24.70)**. Children aged 6–14 and boys were consistently more susceptible. Two problems: no particulate term is reported at all, so the gaseous estimates absorb whatever correlated PM signal exists and cannot be read as pollutant-specific; and a single-hospital catchment in a ten-million-person megacity leaves the denominator undefined and exposed to referral-pattern drift over a decade. Included because the lag structure and the age–sex susceptibility gradient are informative, not because the point estimates are attributable.

doi: [10.3389/fpubh.2026.1885590](https://doi.org/10.3389/fpubh.2026.1885590)

Zhu et al. 2026 Medicine (Baltimore)

Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis

Five years of pneumonia admissions (2015–2019) in an inland Chinese energy city against six pollutants, generalised additive plus distributed-lag nonlinear models. Per 10 $\mu\text{g}/\text{m}^3$: **PM_{2.5} RR 1.004 (0.991–1.017)** and **PM₁₀ 1.004 (0.999–1.010)** — both null — against **NO₂ 1.221 (1.174–1.271)**, **SO₂ 1.035 (1.022–1.048)**, and **CO 1.382 (1.230–1.552)** per 1 mg/m³. Effects peaked at lag 1 for PM_{2.5} and lag 0–7 for the gases; men were more vulnerable to SO₂, women to CO, and children ≤ 14 formed the high-risk group. The single-pollutant framing is the limitation — in a heating-dominated city PM_{2.5}, SO₂, NO₂ and CO share a common combustion source and a common seasonal cycle, and no multi-pollutant model is presented, so the apparent NO₂ dominance is a statement about correlation structure rather than about which pollutant acts. Useful mainly as a demonstration that PM mass can be the null term in its own time-series.

doi: [10.1097/MD.00000000000049962](https://doi.org/10.1097/MD.00000000000049962)

Mechanistic toxicology

2 records

Balu et al. 2026 J Biochem Mol Toxicol

SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction

Langendorff-perfused male Wistar hearts exposed to NIST SRM 1650b heavy-duty diesel particulate at 10, 100 and 300 $\mu\text{g}/\text{mL}$ before 30 min ischemia and 60 min reperfusion. Dose-dependent deterioration of haemodynamics and elevated tissue injury, with increased oxidative stress in tissue and isolated mitochondria, reduced bioenergetic enzyme activity and respiratory efficiency, downregulated *Pgc-1 α* , *Tfam*, *Polg*, *Fis1*, *Mfn1* and *Pink1*, lower mtDNA copy number, and suppressed PI3K/Akt signalling that fell further after the ischemic challenge. The design isolates a direct cardiomyocyte effect cleanly — no autonomic, pulmonary or systemic inflammatory route is available in an ex-vivo preparation — which is both its strength and its ceiling. The concentrations are far above any inhaled dose delivered to myocardium in vivo, and direct coronary perfusion bypasses the alveolar barrier entirely, so this establishes a mechanism that *can* operate, not one that does at ambient exposure.

doi: [10.1002/jbt.71038](https://doi.org/10.1002/jbt.71038)

Pan et al. 2026 Front Pharmacol

Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis

PRISMA systematic review and random-effects meta-analysis of 18 rodent studies, standardised mean differences for nasal mucosal markers. Air pollutant exposure was associated with increased eosinophils and elevated IL-4, IL-5, IL-13 and OVA-specific IgE — a coherent enhancement of type 2 inflammation — with scattered evidence of epithelial barrier loss (reduced ZO-1) and innate activation (IL-1 β , NLRP3). IFN- γ showed no consistent direction. Heterogeneity was substantial and the authors decline to claim a causal pathway. The pooling is the weak point: 18 studies spanning diesel exhaust particulate, PM_{2.5}, PM₁₀ and mixed urban aerosol at uncontrolled doses and durations, in an ovalbumin-sensitisation model whose relationship to human allergic rhinitis is itself contested, cannot yield a transferable effect size. No dose–response meta-regression is attempted, which is the analysis that would have justified the synthesis. Value here is as a marker map for the type 2 axis, not as an effect estimate.

doi: [10.3389/fphar.2026.1870023](https://doi.org/10.3389/fphar.2026.1870023)

Other clinical endpoints

2 records

Zhou et al. 2026 BMJ Open

Environmental exposome study for prostate cancer in elderly men from Western China

580 histologically confirmed localised prostate cancers after radical prostatectomy, matched 1:2 to 1,160 cancer-free controls from the WeEndPd

cohort, with urinary bisphenols and phthalate metabolites by LC-MS/MS and geocoded five-year SO₂ and PM_{2.5}. Highest-quartile five-year SO₂ (9.37–28.79 µg/m³) gave **OR 1.65 (95% CI 1.08–2.72)** against the lowest; MECPP, MEHHP, MEHP, BPA and BPZ were all higher in cases; the WQS mixture of endocrine-disrupting chemicals gave **OR 1.26 (1.20–1.33)**. The PM_{2.5} term is reported but is not the headline. Two structural problems: urinary EDC metabolites have half-lives of hours to days and were measured at or after diagnosis, so they cannot index the aetiologically relevant window — this is reverse-causation-prone by construction; and cases are surgical patients from one tertiary centre, which selects on operability and on health-seeking behaviour that also tracks residential air quality.

doi: [10.1136/bmjopen-2025-110957](https://doi.org/10.1136/bmjopen-2025-110957)

Knapova et al. 2026 Appl Psychol Health Well Being

Daily relationship between air pollution and physical activity in an industrial region

693 adults in the industrial Ostrava region of the Czech Republic, 9,702 person-days, device-measured step counts against daily PM₁₀ over 14 days, multilevel Bayesian models with within-person centring and adjustment for temperature, sunshine and precipitation. On days when PM₁₀ exceeded a participant's own 14-day mean, step counts fell ($b = -0.027$, 95% CI -0.047 to -0.006); **a 1-SD PM₁₀ increase corresponded to 361 fewer steps**. Self-reported tendency to monitor air quality did not moderate the effect, which argues the response is not deliberate avoidance behaviour. Exposure is ambient monitor data assigned to residence, not personal, so the estimate confounds pollution with unmeasured weather and visibility cues; 14 days is short for a within-person design; and non-probabilistic quota sampling limits transport. The finding matters for exposure modelling regardless: **behavioural response makes time-activity patterns endogenous to concentration**, which biases any exposure model that treats mobility as exogenous.

doi: [10.1111/aphw.70186](https://doi.org/10.1111/aphw.70186)

Burden, policy & mitigation

1 record

Qian et al. 2026 J Hazard Mater

Stronger actions are needed to advance PM_{2.5} mitigation and its health benefits under population aging in China

A new daily 1 km PM_{2.5} product for mainland China 2015–2024, used to track concentration, WHO AQG exceedance and population-weighted exposure, with attributable cardiovascular mortality and a decomposition isolating the ageing offset. National PM_{2.5} fell **35.57 → 20.62 µg/m³** (population-weighted -37.35%), and the area above the WHO interim target of 35 µg/m³ collapsed from 51.07% to 1.49% — **yet over 99% of the population still lives above the 5 µg/m³ guideline**. Attributable CVD deaths fell from 2.16 (1.64–2.58) million in 2015 to 1.55 (1.14–1.91) million in 2022, then **rebounded to 1.76 (1.30–2.15) million in 2024**. The share of health benefit cancelled by population ageing rose from around 40% before 2020 to **56.54% in 2024**, exceeding 70% in some subregions. The burden estimates inherit exposure–response functions from external cohorts, so the uncertainty intervals understate structural uncertainty, and the 1 km product's own error is not propagated into the mortality intervals. The demographic-offset framing is the contribution and it generalises beyond China.

doi: [10.1016/j.jhazmat.2026.143101](https://doi.org/10.1016/j.jhazmat.2026.143101)

Occupational & indoor exposure

1 record

Park et al. 2026 Sci Total Environ

Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon

Mock-up chamber characterisation of softwood finishes plus field monitoring in an occupied childcare facility and an unoccupied university classroom — before remodelling, 24 h after, and at four weeks — covering formaldehyde, anthropogenic and biogenic VOCs, particulate matter and total airborne bacteria, with a life-cycle global warming potential assessment. α -Pinene dominated the biogenic emission profile. In the occupied, ventilated building PM and bacterial counts fell or stabilised and most VOCs declined within four weeks; in the unoccupied, poorly ventilated classroom **α -pinene rose from 28.7 to 1713.6 µg/m³ over four weeks** — a 60-fold accumulation. Net GWP in that building fell to 37.7 kgCO₂eq, 9–11 times below non-wood alternatives, on biogenic carbon storage. With $n = 2$ buildings differing simultaneously in occupancy, ventilation and use, those factors cannot be separated, and the PM measurements are reported only qualitatively. The transferable point is the ventilation dependence: a material acceptable in one air-exchange regime is not in another, which argues for continuous indoor monitoring rather than commissioning-time spot checks.

doi: [10.1016/j.scitotenv.2026.182065](https://doi.org/10.1016/j.scitotenv.2026.182065)

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Daoud et al.	Air pollution and suicide mortality in the UK: implications for environmental and mental health policy	<i>Environ Sci Pollut Res Int</i>	Ecological panel	PM _{2.5}	UK
Zhang et al.	Association of PM _{2.5} and its components with cognitive dysfunction among rural older adults	<i>Neurotoxicology</i>	Cross-sectional (multistage)	PM _{2.5} components (BC, OM, NH ₄ , SO ₄ , NO ₃)	China
Reproductive & developmental					
Bello et al.	Prenatal exposure to PM _{2.5} and infant health: evidence from Quebec	<i>Econ Hum Biol</i>	Registry cohort + geospatial	PM _{2.5}	Canada
Pai et al.	Household incense burning and small for gestational age risk in newborn infants	<i>Pediatr Neonatol</i>	Prospective cohort	Incense PM (indoor, unspecified)	Taiwan
Sezer et al.	Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate	<i>Birth Defects Res</i>	Case-control	PM _{2.5} + PM ₁₀ + O ₃	Turkiye
Respiratory & allergic					
Yue et al.	Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou	<i>Front Public Health</i>	Time-series	Traffic gases (NO ₂ , SO ₂)	China
Zhu et al.	Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis	<i>Medicine (Baltimore)</i>	Time-series	PM _{2.5} + PM ₁₀ + 4 gases	China
Mechanistic toxicology					
Balu et al.	SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction	<i>J Biochem Mol Toxicol</i>	Ex vivo perfused organ	Diesel PM (SRM 1650b)	India
Pan et al.	Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis	<i>Front Pharmacol</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀ / DEP (animal)	Global
Occupational & indoor					
Park et al.	Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon	<i>Sci Total Environ</i>	Field measurement	PM (indoor, unspecified)	South Korea
Exposure assessment & modelling					
Chen et al.	Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability	<i>J Hazard Mater</i>	Exposome-wide association	PM _{2.5} constituents + VOCs	USA
Nyoni et al.	Performance of road-traffic-based exposure proxies against personal PM _{2.5} measurements in three Sub-Saharan African countries	<i>Air Qual Atmos Health</i>	Proxy validation vs personal exposure	PM _{2.5} (personal, filtered)	The Gambia / Kenya / Mozambique
Burden, policy & mitigation					
Qian et al.	Stronger actions are needed to advance PM _{2.5} mitigation and its health benefits under population aging in China	<i>J Hazard Mater</i>	Machine-learning model	PM _{2.5}	China
Other clinical endpoints					
Knapova et al.	Daily relationship between air pollution and physical activity in an industrial region	<i>Appl Psychol Health Well Being</i>	Panel study	PM ₁₀	Czech Republic
Zhou et al.	Environmental exposome study for prostate cancer in elderly men from Western China	<i>BMJ Open</i>	Case-control	PM _{2.5} + SO ₂	China

Provenance and method

Entry window **31 July–1 August 2026**. The 31 July issue was a corrections release that carried no harvest, so this window spans two days and is contiguous with the 30 July issue — no gap. Sources: **PubMed** (E-utilities and connector) queried on the [Date - Entry] field along two independent axes, health (13 hits) and instrumentation (9 hits, 2 of them not already on the health axis); **Europe PMC** CREATION_DATE: [2026-07-31 TO 2026-08-01] (2 hits, both new); **arXiv** relevance query with a local recency screen (146 kB of entries, none in window); **OpenAlex** returned HTTP 429 and remains unusable for date-windowed harvesting; **ClinicalTrials.gov** returned no PM or air-pollution registrations or updates in the window. **24 candidates screened, 15 in scope, 9 rejected**, each logged with a reason to `state/rejected.jsonl`. No duplicates against `state/seen.json` (86 PMIDs, 99 DOIs, 95 title signatures at entry).

The rejections are informative and are stated rather than hidden. Four were non-particulate occupational or clinical records that the broad query pulled in: a workplace SO₂ inter-laboratory comparison, a nitrogen-tetroxide push-pull ventilation study, a workplace-air standard-methods commentary, and a tobacco-cessation survey. Three were documents in which air pollution appears only as a named risk factor without data — an ESC consensus statement on inflammation in coronary syndromes, a PCOS review, and a COPD health-literacy survey with no exposure measurement. One was a heat-mortality effect-modification study in which air pollution entered only as a covariate. One — an e-cigarette aerosol transcriptomics study — was excluded as engineered propylene-glycol aerosol rather than ambient or combustion PM; that boundary is a judgement and is recorded here so it can be revisited.

Known defect, second consecutive window: zero sensing or instrumentation records. This is a harvest failure, not a field observation. PubMed does not index *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*; the Consensus connector has no entry-date filter and therefore cannot supply a windowed record (established 30 July); OpenAlex date filtering is paywalled. The planned fix remains a Crossref by-journal feed keyed on `created` for AMT (1867-8548) and ACP (1680-7324). Until it lands, the Sensing subtopic will keep under-counting and the weekly rollup should be read with that bias in mind.

Subtopic, design, particle-metric and geography labels were assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is therefore a judgement, not a controlled vocabulary. Effect estimates in the forest plot are transcribed verbatim from abstracts and use heterogeneous exposure contrasts — see the figure caption. Every DOI was verified against the PMID's own PubMed record and resolved before publication. Content derived from PubMed and Europe PMC metadata; abstracts remain the copyright of their respective publishers.